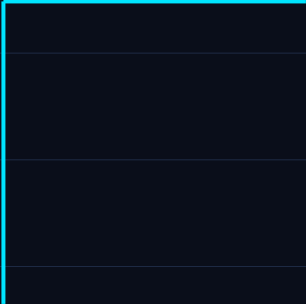




TEXTROID

AI Comic Studio

AMD AI DevMaster Hackathon 2026

Track 1: Multimodal Content Creation Tools

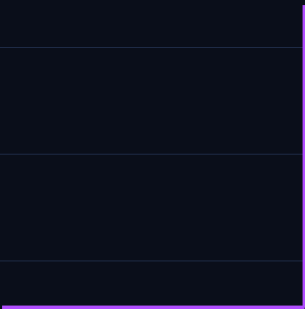
AMD ROCm 7.2

Radeon RX 7900 XTX

Stable Diffusion XL

Gradio 6.x

Python 3.12



Project Overview

Textovid is a multimodal AI application that generates unique, original comic books from text prompts or category selections. It combines large language model storytelling with AI image generation, assembling full comic pages with speech bubbles, narration, sound effects, and a professional title page — all accelerated on AMD Radeon GPUs via ROCm. The system leverages the free Qwen 35B LLM API provided by AMD Radeon Cloud for story generation, and runs Stable Diffusion XL locally on the GPU for high-resolution panel artwork, delivering complete 4K comic pages in a single automated pipeline.

Key Features

Dual Input Modes

Users can either type a free-text story premise or use structured category selectors (genre, sub-genre, art style, theme, mood, length) to guide generation. Both modes produce equally detailed and unique comic pages.

Uniqueness Engine

Every comic is structurally distinct. The randomization engine varies plot structures, character traits, settings, and plot twists. A SHA-256 fingerprint is computed for each generated comic to cryptographically prove uniqueness.

High-Resolution Generation

The image engine uses Stable Diffusion XL in a single-pass configuration at 2048x2880 pixels, producing near-4K panel artwork directly without requiring a separate refiner pipeline.

4K Comic Page Assembly

Generated panels are automatically composed into professional 2480x3508 pixel comic pages (A4 at 210 DPI) with title/cover pages, speech bubbles with quoted dialogue, narration boxes, and sound effect text overlays.

Cyberpunk Gradio UI

The web interface features a sci-fi themed design with animated grid backgrounds, neon accent colors, and HUD-style elements, all built on Gradio 6.x with real-time progress tracking.

Built-in GPU Benchmark

A one-click benchmark measures GPU inference time, VRAM utilization, and throughput, producing a performance score for hackathon evaluation.

System Architecture

Textovid implements a four-stage multimodal pipeline that transforms user input into finished comic pages. Each stage is modular and independently testable, allowing for straightforward extension and maintenance. The pipeline runs entirely on the AMD Radeon Cloud instance, with only the LLM story generation offloaded to AMD's free cloud API (zero GPU cost on the local instance).

Stage	Component	Technology	Execution
1. Input	Uniqueness Engine	CPU / Python random	Local
2. Story	Qwen 35B LLM	AMD Radeon Cloud API	Remote (free)
3. Images	SDXL Pipeline	PyTorch + ROCm 7.2	GPU (Radeon 7900 XTX)
4. Layout	Comic Layout Engine	Pillow (PIL)	Local CPU

Pipeline Detail

The user provides either a free-text story description or selects categories (genre, art style, theme, mood). The Uniqueness Engine randomizes plot structures, character archetypes, and settings to ensure every generation produces a distinct narrative. This randomized context is sent to the Qwen 35B LLM via AMD's free Radeon Cloud API, which returns a structured comic script with panel descriptions, dialogue in quotes, and narration. Each panel description is then passed to the SDXL image engine running locally on the Radeon RX 7900 XTX GPU via ROCm 7.2, generating high-resolution artwork at 2048x2880 pixels in float16 precision. Finally, the Comic Layout Engine assembles all panels onto a 2480x3508 pixel A4-ratio canvas, adding speech bubbles for quoted dialogue, narration boxes, sound effect text, and an optional title page.

AMD ROCm Optimizations

Optimization	Implementation	Impact
Float16 Inference	All SDXL computations use torch.float16 via ROCm	2x speedup, 50% VRAM reduction vs FP32
Attention Slicing	pipeline.enable_attention_slicing() on SDXL	Reduces peak VRAM during generation
Single-Pass Hi-Res	Direct 2048x2880px generation, no refiner	Simpler pipeline, fewer failure points
HIP/ROCm 7.2 Backend	PyTorch 2.11.0 compiled for ROCm 7.2	Full GPU acceleration on Radeon
GPU Benchmarking	Built-in timing + VRAM measurement tool	Quantifiable performance metrics

Technology Stack

Layer	Technology	Version / Details
GPU	AMD Radeon RX 7900 XTX	51.52 GB VRAM on Radeon Cloud
Software Stack	ROCm 7.2 + PyTorch 2.11.0	HIP backend, float16 inference
Image Generation	Stable Diffusion XL	diffusers 0.32+, single-pass 2048px
Language Model	Qwen 35B (A3B)	Free via AMD Radeon Cloud API
Web Framework	Gradio 6.20.0	Cyberpunk sci-fi UI theme
Image Processing	Pillow (PIL)	4K page assembly, speech bubbles, SFX
Python	3.12	Modern type hints, f-strings

Demo Results

A full 4-panel comic page was successfully generated on the AMD Radeon Cloud instance. The generation pipeline completed in approximately 466 seconds for 4 panels at 2048x2880 pixels each, assembled into a final 2480x3508 pixel (15.6 MB PNG) comic page. The GPU reported 44.44 GB of 51.52 GB VRAM in use during generation, demonstrating efficient memory management with float16 inference and attention slicing. The Gradio web interface was successfully exposed via a public gradio.live URL using a manually installed frpc binary, confirming full end-to-end functionality accessible from any browser.

Future Enhancements

The codebase includes a scaffolded video_engine.py module for LTX Video panel animation, which would enable animating individual comic panels into short video clips. This feature is planned as a future enhancement once LTX Video compatibility with ROCm 7.2 and diffusers 0.32+ is validated. Additional planned improvements include multi-page story arcs with character consistency across panels, style transfer from reference images, and batch generation for creating entire comic issues in a single session. The modular architecture supports straightforward integration of these features without requiring changes to the core pipeline.